

Descriptive Statistics and Technical Interpretation

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Introduction

This note works through Task 8 - using descriptive statistics to interpret measurement data from Biochemistry, Electronics, and Mechanical experiments, and understanding what summary numbers reveal, what they hide, and when they can be trusted.

I. Descriptive Statistics and Why They Matter

Descriptive statistics are a set of numbers such as average, spread, range - that summarize a dataset without needing to look at every value.

They come before cleaning or visualization because they act as a first check. A quick look at min/max can catch some bad value (like a negative age or a pH of 20) before you build anything on top of bad data.

A summary only shows what the data looks like, not why. Two very different processes can produce the same mean.

Example: A college reports an average package that looks great - but it's only high because a few students are pulling the number way up. The "typical" student earns far less. The stat is correct; the picture it paints is wrong.

II. Center and Spread

Mean is the sum of values divided by their count. Median is the middle value once sorted. Min and max are the smallest and largest values, and range is their difference. Variance is how far the values typically range from mean. IQR is the spread of the middle 50 percent, from Q1 to Q3.

Center is where the data sits; spread is how far it varies around that point.

The mean gets pulled by one huge or one small number, since it can drag whole average towards it.

The median only cares about the position and not the size.

A high standard deviation means readings are very inconsistent. They are scattered from the average. There is a lot of deviation in the values.

IQR ignores the top and bottom quarters, so it isn't dragged around by skew or outliers the way the mean or range are.

III. Distribution Shape and Sample Size

A distribution shows the full shape of the data. It shows how the values are spread out.

Skewness means more data sits on one side. Clusters hint at subgroups. Gaps mean a range where no data fell. Spread is the overall variability.

Five readings are weaker than five hundred, since one odd value swings a small sample a lot but barely moves a large one.

This is why sample size should be checked before trusting a mean or a pattern on a plot - small samples are fragile and vary a lot if there are any bad values present in the data.

Example: Testing 3 resistors and finding one reads a bit high could look like the whole batch is defective.

IV. Outliers and Domain Context

An outlier is a value that sits well outside where the rest of the data is clustered.

Because some variables genuinely have room for extreme-but-real values.

An outlier is worth investigating if the value is still physically possible; it can be removed outright only if it's impossible for that variable.

A. Biochemistry

A pH dropping from 7.4 to 3.0 could mean contamination, a bad buffer, or a miscalibrated probe.

B. Electronics

A sudden spike in current during an otherwise steady test could be a genuine overload, a brief transient, or an actual short forming in the circuit.

C. Mechanical

An unusually high force reading could mean a stronger sample, a unit mismatch, sensor saturation, or a real shock event.

V. Plot Selection and Interpretation

Histogram - distribution of one numeric variable.

Box plot - centre, spread, and outliers across groups.

Bar chart - one summary value per category.

Scatter plot - relationship between two continuous variables.

Line plot - values in sequence, usually over time.

None of this mean anything without a title, labelled axes, and units - a spike means nothing if you don't know if it's 5 mV or 5 V.

Plot Type	When I'd Reach for It	Example
Histogram	One numeric variable, seeing how its values are distributed	Spread of pH readings collected across a week of buffer prep
Box Plot	Comparing centre, spread, and outliers across several groups at once	Torque readings across three different bolt suppliers
Bar Chart	One summary value (count or average) per category	Average output voltage of four different power-supply units
Scatter Plot	Relationship between two continuous variables	Current vs. voltage across a range of test loads
Line Plot	Values ordered in sequence or time	Reactor temperature logged every 30 seconds during a reaction

VI. Correlation, Causation, and Confounding

Correlation is two variables moving together.

Causation is one variable's change actually producing the others.

Because there are other explanations sitting right next to it - pure coincidence, or some hidden third factor that's driving both variables without either one touching the other directly.

That third factor is a confounding variable - hidden, but influencing both, making them look linked when they aren't.

Example: Lab temperature and reaction yield rise together over a week, but the reagent batch was also replaced midweek - the fresher reagent may be the real driver.

VII. Domain Measurement Interpretation and Comparability

They're completely different physical quantities on different scales - averaging them is like averaging temperature with distance. Whatever number you get doesn't map onto anything real.

Units aren't decoration - a reading of “40” means nothing until you know volts, newtons, or degrees.

Because a comparison only makes sense if you're comparing like with like. Group things properly first, and the resulting averages, comparisons, or plots actually mean something real.

Valid Comparisons

- Voltage on the same circuit, low-load vs. high-load.
- pH of the same solution, before and after treatment.
- Tensile strength across samples, same rig and unit.

Invalid or Misleading Comparisons

- Pressure compared directly against pH.
- Force values in pounds-force averaged with newtons, unconverted.
- Voltage from two boards under different loads, treated as comparable.

Before analyzing, it's worth confirming units, calibration, raw vs. processed status, conditions, and how missing values were handled.

VIII. Final Reflection

How can statistics be technically correct but still scientifically or practically misleading?

A number can be calculated perfectly and still mean nothing, because math never checks if the result makes sense. A mean without a unit is just a digit going nowhere. Average pH with voltage and you get a number that computes but doesn't exist in reality. Small samples make it worse - one odd reading can swing the whole result, with no way to tell if that's the real pattern or just noise. Outliers cut both ways too: drop them blindly and you might lose something real,

keep them blindly and you might distort everything. Missing context - calibration, conditions, load - quietly changes what a number even means. And correlation keeps getting mistaken for causation whenever a hidden factor is really pulling the strings behind both.

IX. Review Questions

1) If mean and median are far apart, what does that suggest about the data?

That indicates the presence of outliers in the data. Outlier pulls the mean value to one side whereas the median value remains same since it only cares about the middle position.

2) Why is standard deviation not enough without knowing the unit and context?

For suppose the standard deviation is simply “10”. Simply 10 doesn’t make any sense. There should be a unit and a context showing what that value actually means.

3) Is every outlier wrong? Give one case where an outlier is valuable.

No. A student getting 99 marks in the exam where all other students scored below 50 is not wrong. That is the real result not an error.

4) Why is 105 percent attendance a valid data type but an invalid value?

105 is a normal number, so the data type is fine, but attendance is capped at 100 percent, so the value itself is out of range.

5) Why is an average of pH, voltage, current, and force meaningless?

They are all measuring different quantities and averaging out different kind of things will give a number which doesn’t correspond to anything real.

6) Which is more robust to outliers: mean or median? Why?

Median only cares about where that value exists and ranks. Its not about how big or small the number is.

7) What does a box plot show that a bar chart does not?

A box plot shows median, IQR, and outliers. A bar chart just gives you one number per category and tells you nothing about how the data inside that category is spread out.

8) If attendance and score have positive correlation, what can you conclude and what can you not conclude?

You can say they move together. What you can’t conclude is that attendance is causing the better scores.

9) What is one example of a statistically correct statement that is scientifically useless?

Something like "the average of pH, voltage, and force is 5.2." The arithmetic value is fine, but the number doesn't represent anything physical since those quantities can't be meaningfully combined.

10) What extra metadata would you ask for before analysing Biochemistry, Electronics, or Mechanical measurements?

The unit each reading was taken in, whether it's raw or already processed, how the instrument was calibrated, the condition it was recorded under, and how any missing or bad readings got handled.

References

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